Proposal: A Smart Assistant for JUST University Regulations

Team Project — Phase 1: Problem and Data · Building LLM Systems: From Models to Production · For Data Science undergraduates, Faculty of Computer and Information Technology, Jordan University of Science and Technology (Irbid)

1. Team and roles (Phase 1)

Member	Phase 1 role	Responsible for
Hamza Moeen Rasheed Issa	Product lead + Evaluation lead	The user and problem section; writing and checking the 30 test questions, the expected answers and the scoring method.
Mohammad Ziad Abdelmajeed Alrousan	Data lead	Collecting and documenting the source documents (sources list, links); checking Arabic text extraction and OCR numbers; the data card.
Asem Zakaria Alfreeh	Engineering lead	The GitHub repository, the planned architecture, and checking that it fits free compute (Colab/Kaggle, free HF and NVIDIA APIs).

Roles rotate at the start of each phase, so every member will hold every role by the end of the project.

2. User and problem

User: undergraduate Data Science students in the Faculty of Computer and Information Technology at JUST.
Task: finding the correct academic rule (absence limits, withdrawal rules, excuses, exams, GPA and graduation rules, course prerequisites).

Today: the answers are scattered across the degree regulation, the study plan, the academic calendar and many separate official PDFs. Students search the documents by hand or ask friends, and can get wrong information about rules that affect their fees and grades.

Evidence so far:

- **Fragmentation:** the university's regulations pages list dozens of separate documents. We reviewed about 20 relevant ones. A simple question such as "how many absences before I am barred?" needs the 17-page degree regulation *and* the calendar's lecture counts.
- **Inconsistency:** two official documents give different deadlines for a medical excuse (2 working days to the health center vs. 2 weeks to the dean).
- **Accessibility:** about 8 of the ~20 PDFs are scanned images or have broken Arabic text, so they cannot be searched or copied.
- **Test questions:** we wrote 30 questions that cover what students typically need (attendance, withdrawal, excuses, grading, graduation, prerequisites) directly from the documents. The expected answers were written before running any model.

3. What the system will do

A chat assistant that answers students' questions in Arabic (and English) about regulations and prerequisites, and cites the source document and article with every answer. If the question needs a personal record or an administrative decision, it says "I don't know" and points to the right office.

1. Input: "How many absences bar me from a course?"
Answer: More than 20% of the course hours bars you from all later exams in that course. For a course that meets 3 days a week (45 lectures) that is the 10th absence. *Source: Degree Regulation No. 1/1987 (amended), Art. 7/b.*

2. Input: "Until which week can I withdraw from a theory course?"
Answer: Up to week 14 of the semester (week 13 for practical courses). After the first week the course stays on your record as "withdrawn" and fees are not credited. *Source: Degree Regulation, Art. 12.*

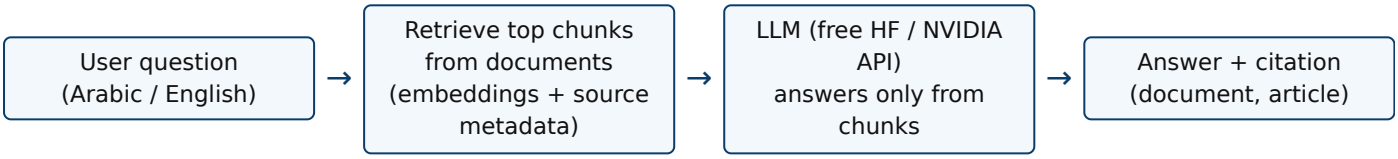
3. Input: "What is my GPA, and will I graduate on time?"
Answer: "I don't know. I have no access to personal records; check the registration portal or your academic advisor."

4. Data

Source	Content	Status
Degree Regulation No. 1/1987 (amended)	Study load, absence, withdrawal, postponement, GPA, probation, graduation	In hand (17 pages)
Exam rules, medical excuses, student discipline 2024, student conduct code, weather suspension, schedule rules, fees, loans, level exams, change of major	Student-facing procedures	In hand (about 18 PDFs)
Academic calendar 2026-2027, 1st semester	Registration, withdrawal and exam periods; lecture counts	In hand (3 pages); dates kept out of the Phase 1 test set
Data Science study plan (prerequisite chart, 2022)	Course prerequisites (pass or study type)	In hand (image); the official current plan will replace it

All sources are published for students on the university website. Language: Arabic. Details and samples are in the data card and the repository.

5. Planned architecture (first version)



If retrieval confidence is low or the question is personal, the system answers "I don't know". **Planned for later phases:** convert the prerequisite chart to a table and add an agent tool that checks a student's course-registration prerequisites (RAG + agent), compared against the simpler RAG version on the same test set.

6. How we will measure success

- **Accuracy** on the 30-question test set (answer matches the expected answer; reviewed manually, with an LLM judge as a helper).
- **Citation correctness:** share of answers that cite the right document and article.
- **Correct refusals:** the system must refuse or say "I don't know" on the 3 out-of-scope questions (and similar ones added later).

The test set has 30 questions: 15 easy, 10 medium, 5 hard. All expected answers were written from the documents before running any model.

7. Risks and backup plan

- **Arabic PDF extraction:** scanned or broken PDFs need OCR and the numbers may be wrong. *Backup:* verify numbers manually and keep key tables (grades, deadlines) as hand-checked CSV.
- **Conflicting or outdated documents** (e.g., excuse deadlines; the 2022 study plan; a calendar for one semester). *Backup:* store the source year with each chunk; answer with both rules and cite them; use the official current plan.
- **A wrong answer about a rule can cost fees or grades.** *Backup:* always show the source; label the assistant as guidance, not an official decision.
- **Arabic quality of free models.** *Backup:* test in Phase 2 and compare with a closed model for development only.
- **Scope and free compute.** *Backup:* the first version covers attendance, withdrawal, excuses, exams, grading and prerequisites only; other documents are added later.